

Mobile application for pine variety classification in an Oxapampa nursery using CNN

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Abstract—Peru has large expanses of reforested hectares along its coast, with pine being one of the two most commonly used species for this practice. These plants originate from forest nurseries, which are centers for the reproduction and care of plants in their early stages of life before being transplanted to the field. A crucial stage in pine production is the classification of seeds by variety, which is fundamental for plant development. This influences the planting destination, subsequent care, and the transplant environment. In this article, we present a mobile application (app) for classifying pine seeds by variety. The app uses a model based on Convolutional Neural Networks (CNN) designed to distinguish and predict among the *Tecunumanii*, *Oocarpa*, and *Pseudostrobus* pine varieties. The mobile application was built in three phases: Data Selection, Model Training, and Development and Implementation. The model achieved an accuracy of 90.32% in classifying pine seeds, demonstrating effectiveness and potential as a useful tool in managing the seed classification process in forest nurseries.

Keywords—*Seeds classification, Pine nuts classification, Artificial intelligence, Convolutional Neural Network, Machine Learning*

I. INTRODUCTION

The reforestation of pines in Peru is an important practice in the forestry sector, but the seed production process lacks adequate management.

The pine seed production process in Peru faces challenges that jeopardize the quality of forest plantations and the sustainability of the resource. It was found that 83% of seeds are collected informally and without control from natural forests, without knowing the genetic value or the actual species, directly affecting the quality and variety of seeds distributed in the country [1]. Additionally, poor seed classification by variety can lead to seed mixing during production, negatively impacting seed purity and, consequently, market stability and agricultural yields [2]. Pine seeds have high morphological similarity, making variety classification difficult [3]. Moreover, due to the lack of traceability information in distribution flows, the origins are uncertain, and the accuracy of variety is null [4]. These difficulties generate uncertainty about the quality and legitimacy of seeds, negatively impacting forest conservation [5] and affecting economic benefits in the forestry sector economy [6].

There are many studies on using artificial intelligence, machine learning, and deep learning techniques for seed classification. Among the findings, [4] presents a study in Turkey where a mobile application based on CNN architectures was developed for seed variety selection,

achieving up to 99% accuracy in validation. Another study [7] compared various classification methods using CNN, achieving a maximum accuracy of 99.9% and recommending the development of mobile applications for this purpose. However, Peru lacks solutions to classify pine seeds by the varieties produced and planted in the country. Therefore, our proposal is to develop a mobile application for seed classification to determine three species of pine seeds (*Tecunumanii*, *Oocarpa*, *Pseudostrobus*) using CNNs with artificial intelligence libraries and Python data analysis for the classification model and React Native for mobile application development.

The development will occur in three phases: (1) data selection, (2) model training, and (3) development and implementation of the app.

This work is organized into sections as follows: Section 2 presents an analysis of related articles, Section 3 shows the proposed model, Section 4 addresses the validation, Section 5 presents the results, and finally, Section 6 outlines the conclusions and future work.

II. RELATED WORKS

This section presents the literature review on seed classification using CNNs in mobile applications. It consists of three phases: planning, development, and findings. During the planning phase, the following will be carried out: (1) formulation of research questions, (2) selection of scientific databases, (3) definition of keywords, and (4) definition of inclusion and exclusion criteria. The defined research questions were: Q1: What techniques are currently used by researchers to classify seeds using CNNs? Q2: What are the phases of the proposed techniques? Q3: What algorithms do the found approaches use to build a seed classification solution? Q4: What technologies were found for the development of mobile applications for seed classification?

The defined keywords were: 'seed classification,' 'convolutional neural networks,' 'mobile application,' 'CNN,' 'image classification,' and 'species recognition.' The selected databases for the article search were Scopus, MDPI, and Web of Science.

TABLE I. RESEARCH QUESTIONS ARTICLES

Category	References	Quantity
Technique (Q1)	[2] [8] [9] [10] [11] [12] [13] [3] [14] [15] [16] [17]	12
Phase (Q2)	[9] [11] [14]	3
Algorithms (Q3)	[18] [11] [9] [14] [10] [3] [2]	8
Mobile Applications (Q4)	[4] [14] [14] [9]	3

Regarding techniques, we found 3: Computer Vision [2] [8], CNN(Convolutional Neural Networks) [9] [10] [11] [2] [13] [14] [16] [3] and NIR(near-infrared) [3] [17] CNNs are considered the most important for our study, as they involve the use of non-invasive techniques that can be adapted to mobile developments due to the wide variety of architectures available. This will help define the technique to be used in the classification of pine seeds in the application to be developed.

Regarding the phases, we found 2 sequences. In the first sequence, the phases converge into three stages for seed classification: (1) dataset formation, (2) training, and (3) testing [9] [14]. While in the second sequence, these three phases are supplemented by an additional phase called "Image Feature Extraction" [11]

Regarding the algorithms, we found 5: kNN (k-Nearest Neighbors) [18], DT(Decision Tree) [18] [3], Naïve Bayes [18], Bayes Net [11], CNN [9] [14] [10] [3] [2]. The most prominent and by far the most studied with better results for seed classification through images is CNN. This provides us with a clear path for selecting the algorithm to implement in the selection of pine seeds by variety.

Regarding mobile applications, we found 2: the first one in Python [4] and the second in Java [14], Both articles implement the Machine Learning library called TensorFlow as part of their implementation.

Our analysis of related works concludes that the use of Convolutional Neural Networks (CNN) is closely related to seed classification in general. Architectural variations of this model, like the base model, achieve a high accuracy percentage of over 90% in seed classification. On the other hand, a modern approach will be presented for the development of the application; scientific articles related to the development of mobile applications for seed classification do not provide definitive results on which technology is best suited for this case [14] [9] [4].

III. PROPOSED MODEL

Our proposal is to develop a mobile application based on a Convolutional Neural Network (CNN) model of our own design. The model used by our application was selected based on the following validation of comparison results of models found in related works tested under the same scenario and data.

TABLE II. CNN MODEL ACCURACY PERFORMANCE COMPARISON

Model	Test Accuracy %	Test Loss
Own Custom CNN	97.3125	0.14804404973983765

ResNet50	96.9785	0.1754242479801178
VGG16	92.1875	0.6280940771102905

The system was implemented in 4 phases: collection of pine seed images, development and training of the classification model, design, development, and implementation of the mobile application, and validation of results. The mobile application is developed under a REST architecture consisting of a Frontend client to display data to the user, backend logic responsible for processing images, predicting the variety of seeds, providing that information to the Frontend, and storing it in the database as required. In addition to seed classification functionality, the application provides a supplier management system, worker management, and records of classifications performed.

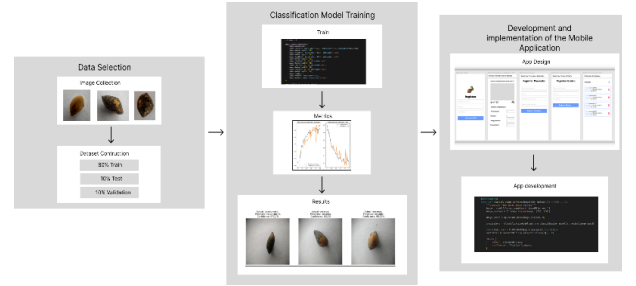


Fig 1 Proposed Mobile Application for Pine Seed Classification

A. Phases of Proposed Model

1) *Phase 1. Data Selection:* This phase aims to collect a diverse set of images of pine seeds from varieties *Pseudostrobus*, *Oocarpa*, and *Tecunumanii* for use in training, testing, and validating the CNN model. It is divided into two sub-phases:

a) *Image Collection:* Seeds were collected through photographs of three varieties of pine seeds with high definition using a 100x magnification lens. The collected images belong to the varieties *Oocarpa*, *Tecunumanii*, and *Pseudostrobus*. It is worth mentioning that all images were captured against a white background, using natural light without any artificial lighting assistance.

b) *Dataset Construction:* Here, the collected images were distributed into three-folders: 80% of the images for training, 10% for testing, and 10% for validation. Each folder has three subfolders named after the seeds to be classified (*Oocarpa*, *Tecunumanii*, and *Pseudostrobus*), which contain the corresponding images of the seed variety.

TABLE III. DATASET DISTRIBUTION

Percentage of images by tags	Tags	Varieties		
80%	Training	Tecunu- manii	Oocarpa	Pseudos- trobus
10%	Evidence	Tecunu- manii	Oocarpa	Pseudos- trobus
10%	Validation	Tecunu- manii	Oocarpa	Pseudos- trobus

2) *Phase 2. Model Training:* This phase aims to develop a pine seed classification model based on the base

architecture of CNN using the collected images. It is divided into three sub-phases.

a) *Training*: This stage marked the beginning of the development of the Machine Learning model, specifically Convolutional Neural Networks (CNN). To carry out this process, the Python programming language was used along with specialized li-braries in data science and artificial intelligence: Matplotlib, Keras, and TensorFlow.

```
BATCH_SIZE = 32
IMAGE_SIZE = 256
CHANNELS=3
EPOCHS=50
```

Fig 2. Definition of Image Parameters for CNN Model Training

The constants shown at the top determine the image size, the number of channels (RGB), batch size, and number of epochs (the number of times the model will see the complete dataset).

```
resize_and_rescale = tf.keras.Sequential([
    layers.experimental.preprocessing.Resizing(IMAGE_SIZE, IMAGE_SIZE),
    layers.experimental.preprocessing.Rescaling(1./255),
])
```

Fig 3. Image resizing function

These layers resize the images to IMAGE_SIZE x IMAGE_SIZE and scale their values between 0 and 1. Data augmentation techniques are also added to improve the model's generalization.

```
input_shape = (BATCH_SIZE, IMAGE_SIZE, IMAGE_SIZE, CHANNELS)
n_classes = 3

model = models.Sequential([
    resize_and_rescale,
    layers.Conv2D(32, kernel_size = (3,3), activation='relu', input_shape=input_shape),
    layers.MaxPooling2D((2, 2)),
    layers.Conv2D(64, kernel_size = (3,3), activation='relu'),
    layers.MaxPooling2D((2, 2)),
    layers.Conv2D(64, kernel_size = (3,3), activation='relu'),
    layers.MaxPooling2D((2, 2)),
    layers.Conv2D(64, (3, 3), activation='relu'),
    layers.MaxPooling2D((2, 2)),
    layers.Conv2D(64, (3, 3), activation='relu'),
    layers.MaxPooling2D((2, 2)),
    layers.Conv2D(64, (3, 3), activation='relu'),
    layers.MaxPooling2D((2, 2)),
    layers.Flatten(),
    layers.Dense(64, activation='relu'),
    layers.Dense(n_classes, activation='softmax'),
])

model.build(input_shape=input_shape)
```

Fig 4. CNN Model

This process defines and constructs a convolutional neural network model for image classification, which includes multiple convolutional and max-pooling layers for feature extraction, followed by dense layers for final classification. Below is the entire flow of the model training.

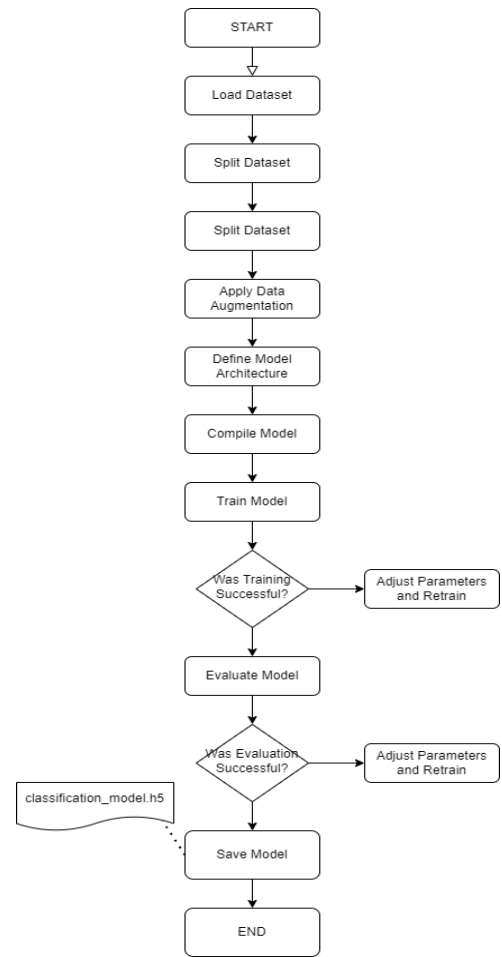


Fig 5. Training Model Flow

b) *Metrics*. Two metrics were used for the developed neural networks: "accuracy" and "loss," which refer to the accuracy value in seed classification and the loss value respectively.

The "accuracy" is calculated by comparing the model's predictions with the true labels in the training set.

$$accuracy = \frac{\text{Number of correct predictions}}{\text{Total number of examples}}$$

While the 'loss' is calculated by applying the loss function to the model's predictions on the training set. It is defined as:

$$L(y, \hat{y}) = - \sum_{i=1}^C y_i \log(\hat{y}_i) \quad (1)$$

Where in (1):

- y is the vector of true labels
- $(\hat{y}_i)^{\wedge}$ is the vector of predicted probabilities by the model
- C is the number of classes

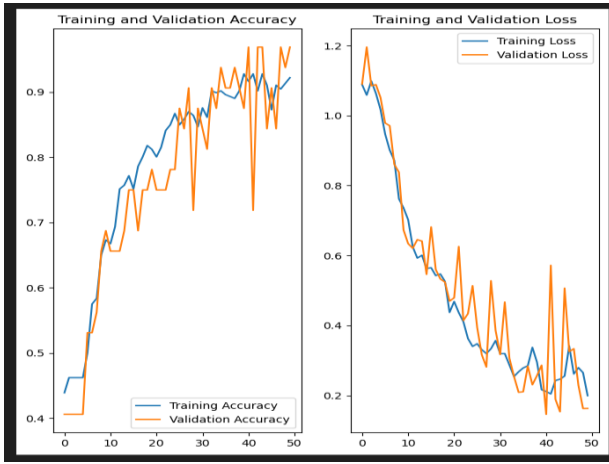


Fig 6. Metric Values During Training

c) *Results:* As a result of this training phase, generous statistics above 90% were obtained with the collected images.

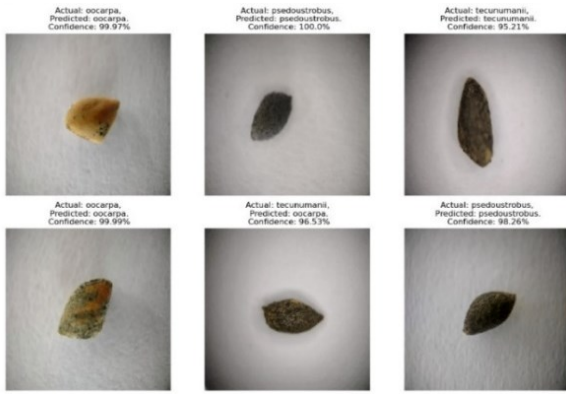


Fig 7. Accuracy Result

Result of the confusion matrix that allows to visualize the classification errors and determine which seed varieties are being confused most frequently.

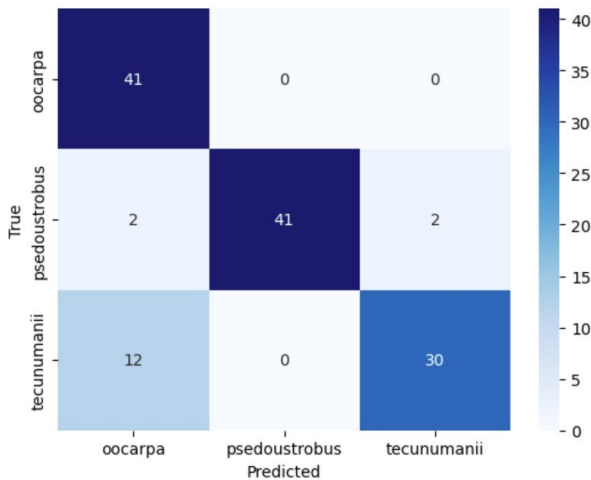


Fig 8. Confusion Matrix

Similarly, a file with the extension .h5 was generated. This file encapsulates the CNN model, including the parameters and weights learned during training. This type of

file (.h5) is commonly used to store trained models in Python, facilitating their subsequent loading and use in different applications or environments. This .h5 file constitutes a tangible document representing the final product of the development and training phase of our model, ready to be implemented in the next phase.

3) *Phase 3. Development and Implementation of The Application:* This phase aims to design, develop, and implement an intuitive mobile application that allows users to capture and classify im-ages of pine seeds in real time using the camera of a mobile device. It integrates the classification model efficiently to pro-vide a smooth and accurate user experience. It consists of three sub-phases

a) *Mobile Application Design:* The design was created using the interface design application Figma and consists of 6 interfaces dedicated to user authentication, seed classification management, and seed supplier management.



Fig 9. Prototypes of Proposed Mobile Application

b) *Mobile Application Development:* The development of the application was carried out using functional programming and a component-based architecture supported by the TypeScript programming language and the React Native cross-platform mobile development library with Expo. The main business logic implemented in the mobile application pertains to seed classification. When creating a pine seed classification session, users can take photographs that the application processes through the CNN model, which is consumed and processed in the application logic developed in Fast API using Python. The application then provides a result that is stored in the database. The process of seed classification through photographs re-peats until the user closes the classification session.

```
@router.post("/predict")
async def classify_image(image: UploadFile = File(...)):
    classification = await ClassificationService.classify_image_service(image)
    return classification
```

Fig 10. Seed Classification API in Fast API

```
@staticmethod
async def classify_image_service(imageFile: UploadFile = File(...)):
    """Classify Pine Seeds Image Service"""
    image = read_file_as_image(await imageFile.read())
    image_resized = tf.image.resize(image, (256, 256))

    image_batch = np.expand_dims(image_resized, 0)

    predictions = ClassificationModel.get_cnn_classification_model().predict(image_batch)
    predicted_class = CLASS_NAMES[np.argmax(predictions[0])]
    confidence = round(100 * (np.max(predictions[0])), 2)

    return {
        'class': predicted_class,
        'confidence': float(confidence),
    }
```

Fig 11. Service Consuming the Classification Model in Fast API


```

export const getSeedClassification = async (photo: CameraCapturedPicture) => {
  const formData: any = new FormData();

  formData.append("image", {
    uri: photo.uri,
    name: "image",
    type: "image/jpeg",
  });

  const response = await axios.post<ClassificationPredictResult>({
    url: `${API_BASE_URL}/classifications/predict`,
    data: formData,
    headers: {
      Accept: "application/json",
      "Content-Type": "multipart/form-data",
    },
  });

  return response;
};

```

Fig 12. Consuming the Pine Seed Classification API from React Native

IV. VALIDATION

To validate the pine seed classification application by variety, three experiments were conducted. Experiment 1 aimed to determine if the application correctly classifies Tecunumanii pine seeds. Experiment 2 aimed to ensure correct classification of Oocarpa pine seeds, while Experiment 3 focused on verifying the correct classification of Pseudostrobus pine seeds.

For these experiments, the following setups were used: 50 grams of Tecunumanii pine seeds (Container 1), 50 grams of Oocarpa pine seeds (Container 2), and 50 grams of Pseudostrobus pine seeds (Container 3).



Fig 13. Pine Seed of the Tecunumanii, Oocarpa and Pseudostrobus Varieties

For all three experiments, the following tools were used: An An-droid smartphone, a x100 magnification lens, and a white card-board canvas.

A. *Experiments*: All experiments will be explained in the following paragraph, as they share the same experimental phases with the difference that they aim to achieve different objectives (determining different varieties of pine seeds).

1) *Software Setup*: For the validation of the proposed application, the developed application was installed on a Moto g(30) smartphone.

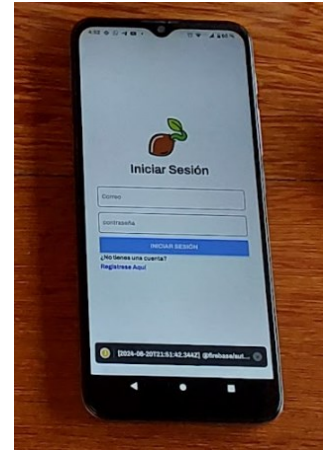


Fig 14. Capturing an image through the proposed application

2) *Hardware Setup*: To capture the morphological details of the pine seeds, a x100 magnification lens, model 100x Microscope by APEXEL, was used.



Fig 15. X100 Magnification Lens



Fig 16. Smartphone with x100 Magnification Lens

3) *Scenario Setup*: In a naturally lit space, a white canvas was placed. This served as a background to rest the pine seeds and perform the classification by capturing photographs with the smartphone camera using the application.

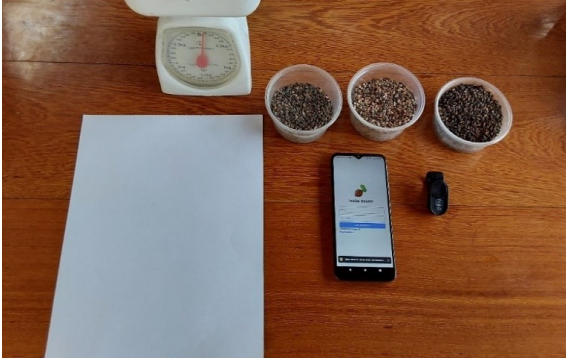


Fig 17. Validation Scenario

4) *Classification*: Upon opening the application, the classification login functionality was used. This allows for the classification of seeds by variety through photographs taken via the application, storing the information of the results. The 50 grams of seeds determined in each scenario were used for each experiment.

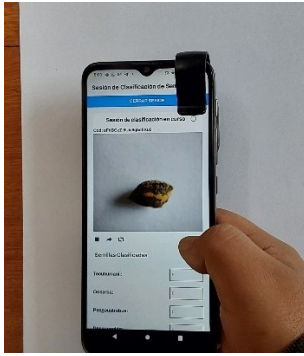


Fig 18. Capturing an image through the proposed application

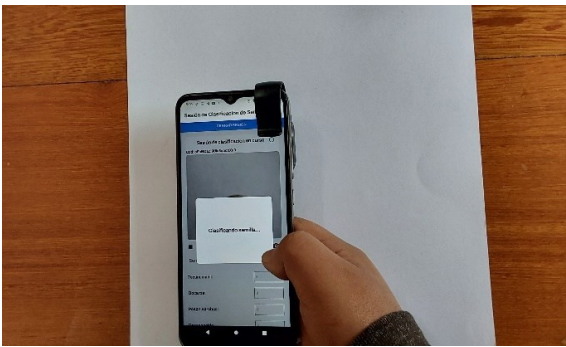


Fig 19. Seed Classification through the application

The classification information was stored in the Firestore NoSQL database, and a summary of these results could be viewed through the "Classification Details" view within the application.

Only when the result of the corresponded to the variety being analyzed was it considered that the seed variety was correctly classified by the application. For example, in Experiment 1, if processing the photograph of scenario 1 resulted in the application indicating that the seed was of the Oocarpa variety, this was considered a failed result. However, if the result indicated that the seed was of the Tecunumanii variety, it was determined to be a successful

result, as scenario 1 contained only Tecunumanii seeds. The same logic was applied to the other experiments.

B. Validation of Experiments: The values obtained from the experiments were considered to determine the effectiveness of the application as a tool for classifying pine seeds by variety. The following formula was used to determine the effectiveness.

$$r = \frac{a}{p} * 100\% \quad (2)$$

Where in (2):

r : result

p : number of seeds evaluated

a : number of successful seeds

V. RESULTS

This section presents the results obtained from the test conducted under the scenario described at the end of the previous section.

In Experiment 1 (classification of 50 grams of Tecunumanii seeds), tests were conducted with a total of 3,227 seed units, and the application yielded a confidence level of 86.98%, correctly identifying the variety in a total of 2,807 units.

TABLE IV. RESULTS OF THE FIRST EXPERIMENT WITH TECUNUMANII SEEDS

Seed Variety	Number of Seeds Evaluated	Number of Seeds Correctly Classified	Accuracy Percentage
Tecunumanii	3,227	2,807	86.98%

In Experiment 2 (classification of 50 grams of Oocarpa seeds), tests were conducted with a total of 3,494 seed units, and the application yielded a confidence level of 95.99%, correctly identifying the variety in a total of 3,354 units.

TABLE V. RESULTS OF THE SECOND EXPERIMENT WITH OOCARPA SEEDS

Seed Variety	Number of Seeds Evaluated	Number of Seeds Correctly Classified	Accuracy Percentage
Oocarpa	3,494	3,354	95.99%

In Experiment 3 (classification of 50 grams of Pseudostrobus seeds), tests were conducted with a total of 3,659 seed units. The application achieved a confidence level of 87.97%, correctly identifying the variety in a total of 3,219 units.

TABLE VI. RESULTS OF THE THIRD EXPERIMENT WITH PSEUDOSTROBUS SEEDS

<i>Seed Variety</i>	<i>Number of Seeds Evaluated</i>	<i>Number of Seeds Correctly Classified</i>	<i>Accuracy Percentage</i>
Pseudostrobus	3,659	3,219	87.97%

The accuracy percentage corresponds to the effectiveness of the model's use through the application concerning the seed variety. We can see that the accuracy percentage is generally high but significantly different for the Tecunumanii variety compared to the Oocarpa variety. However, it is still much higher than the manual selection result provided by the forest nursery, which has an average accuracy percentage of 75.4%.

TABLE VII.

<i>Number of Seeds Evaluated</i>	<i>Number of Seeds Correctly Classified</i>	<i>Accuracy Percentage</i>
10,380	9,380	90.32%

VI. CONCLUSIONS AND FUTURE WORK

The research allowed us to achieve positive results in classification, achieving accuracy rates of 86.98% for the first scenario, 95.99% for the second scenario, and 87.97% for the third and final scenario. This resulted in an average classification accuracy of 90.32% through the application. This means that the application has a high level of confidence and can be used to support the seed selection process in a pine seedling production environment of the Tecunumanii, Pseudostrobus and Oocarpa varieties, especially in forest nurseries.

Various studies were found in the literature for seed classification in general and pine seeds specifically. However, none of them presented a classification tool for Tecunumanii, Pseudostrobus, and Oocarpa pine seed varieties.

This study proposed a mobile application for pine seed classification for the three most used varieties in Peru. The application was developed in three phases: (1) data selection, (2) model training, and (3) application development and implementation.

Three experiments were conducted to validate the application at a forest nursery in the province of Oxapampa, Pasco department, Peru. Experiment 1 validated the classification of 50g of Tecunumanii pine seeds using the application. Experiment 2 validated the classification of 50g of Oocarpa pine seeds, and Experiment 3 did the same for Pseudostrobus pine seeds.

As future work, we encourage strengthening the classification model by training it with images of pine seeds in different scenarios, including various types of light, including artificial ones, different backgrounds on which the seed is placed, and various varieties of pine seeds not used in the research, such as Pinus Patula, Caribaea, Radiata, among others. We also want to expand the test scenario to the regions of Cusco and Huancavelica, places where there are pine production centers. This will make the application

a more robust system that can be used in different situations not included in the study.

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